PCMRITMS - Phase-Controlled Multi-Ring Inertial Torque Modulation System

Concept Whitepaper v1.0

Project Phoenix

Document status: Conceptual / simulation-validated - no hardware prototype measured

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Disclaimer: All performance figures in this document are engineering projections from physics-based computer simulation unless explicitly marked measured. The 75-85% round-trip efficiency target is optimistic and unvalidated. Prefix external claims with ?simulation shows...? Hardware validation is required before any commercial decision.

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1. What It Is

The Phase-Controlled Multi-Ring Inertial Torque Modulation System (PCMRITMS) is an electromechanical buffer placed between a primary energy source (battery, traction motor, or engine-generator) and the driveline. It stores, shapes, and releases rotational energy on millisecond-to-second timescales using **multiple mechanically independent coaxial inertial rotors**, each with its own motor-generator and magnetic bearings.

It is an active inertial buffer and torque shaper - not a replacement for the primary power source.

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2. Reference Architecture

- Independent coaxial rotors (nested rings) whose speeds are not kinematically locked.
- Each rotor: dedicated magnetic-bearing set + motor-generator stator.
- Reaction torques routed to a torque-summing stage on the central output shaft.
- Electrical path: DC bus -> per-rotor inverters -> MG windings; encoders close the loop per rotor.

Torque-summing options

Option	Description	Status
A. Epicyclic summing	Each rotor MG stator grounded to carrier; planet/carrier torque sums to output	Primary reference (~1-3% gear loss per stage)
B. Direct MG reaction	MG stator bolted to summing housing	Alternative
C. Magnetic torque coupling	Non-contact rotor to output	Research stage

Selected reference: Option A (epicyclic).

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3. Operational Modes

Mode	Objective	Behavior
Torque boost	Short acceleration pulse	Rotors discharge stored KE; phases aligned for constructive reaction torque
Energy recovery	Braking / downhill	Rotors accelerate, absorbing driveline energy
Smoothing	Reduce ripple	Phases adjusted to cancel oscillatory components
Decoupling	Protect battery/motor	Primary source held near efficient point; buffer supplies transient deficit

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4. Representative Sizing (Phase 1, illustrative)

Parameter	Value
Number of rotors	3 (expandable to 4-6)
Per-rotor inertia	0.10-0.15 kg·m²
Operating speed (mean)	~7,600 rpm (800 rad/s)
Stored energy (total)	118 kJ (0.118 MJ)
Per-rotor MG continuous rating	30-35 kW
Pack electrical rating (3 rotors)	90-105 kW
Continuous discharge (vehicle model)	90 kW
Brief burst (rotor-coupled, vehicle model)	140 kW
Target peak torque boost (brief)	+35-50% vs motor-only baseline
Boost duration (typical)	0.2-0.5 s (energy-limited)
Energy at full 90 kW discharge	~1.3 s until depleted
Target round-trip efficiency	75-85% (unvalidated)
Modelled round-trip efficiency	80%
Envelope (automotive auxiliary)	~40-60 L, 45-70 kg

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5. Working Principle

Per rotor i, electromagnetic torque accelerates the rotor; the reaction torque on the stator/summing stage is the useful output contribution:

tau_react,i = -I_i x alpha_i

Total output torque:

tau_out = tau_primary + Sum tau_react,i

The controller selects per-rotor amplitude, modulation frequency, and phase subject to DC-bus voltage/current limits, maximum speed and acceleration, minimum stored energy, and thermal limits.

Hard energy constraint

Peak output torque cannot exceed what stored kinetic energy and electrical input support.

Sustained torque multiplication for multiple seconds is not physically realistic at vehicle packaging. The system delivers brief, shaped pulses and driveline-quality improvement - not continuous overpowering of the traction motor.

Gyroscopic management

Counter-rotating rotor pairs (opposite mean angular momentum) reduce net angular momentum versus a single large flywheel while each member still phase-modulates for torque shaping.

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6. Headline Simulation Result

Lumped-parameter model (3 rotors, losses neglected in torque peak):

Rotor parameter	Value
Inertias	0.12, 0.15, 0.10 kg·m²
Mean speed	800 rad/s
Speed modulation amplitude	80 rad/s
Modulation frequency	18 rad/s
Phase separation	120 deg
Fixed primary torque	180 N·m

Metric	Value
Baseline torque (motor only)	180 N·m
Peak combined torque	242.8 N·m
Peak torque increase	+34.9% (brief)
Total stored kinetic energy	0.118 MJ
Rotor surge power (vehicle integration model)	50.2 kW
Brief burst rating on DC bus	140 kW

Note: instantaneous electrical demand in the unconstrained model reaches ~305 kW; production design caps per-inverter ratings (~30-35 kW continuous per rotor), shortening the real boost window.

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7. Integration with Project Phoenix Powertrain

When coupled to the ATPE + battery series hybrid on an 800 V DC bus:

Role	PCMRITMS contribution
Covers ATPE tier-switch gap (~200-400 ms)	Seamless transient supply
Smooths free-piston electrical ripple	Rotor filtering
Absorbs acceleration spikes	118 kJ sprint reserve
Refill between spikes	ATPE surplus charges rotors to operating speed

Honest finding from vehicle simulation: smarter timing of flywheel discharge makes **no measurable difference on normal driving - the buffer is energy-limited**, not power-limited. The sprint reserve depletes before burst power is exhausted.

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8. Simulation Limitations

- No multi-body epicyclic dynamics or gear-mesh stiffness
- No inverter saturation or thermal derating in headline torque model
- No vehicle longitudinal model in standalone rotor validation
- Round-trip efficiency used in vehicle model (80%) is within the stated 75-85% target band but is not measured

9. Safety and Integration

- Safety: high-speed rotors require burst containment (SAE practices), rotor-burst sensors, safe shutdown on bearing or inverter fault.

- Integration: parallel (torque-additive) to driveline or series-hybrid node; OEM collaboration required for packaging, NVH, and fault diagnostics.

- Functional safety: ISO 26262 ASIL assignment to be determined.

10. Development Roadmap

Phase	Timeline	Deliverable
1 - Modelling	0-6 months	Drive-cycle energy benefit with loss models
2 - Single-rotor lab	6-12 months	One MG + magnetic bearing + dynamometer
3 - Multi-rotor	12-24 months	Phase control; summing stage; gyroscopic pair
4 - Vehicle demo	24-36 months	Fleet pilot; durability and containment test

11. Summary

PCMRITMS delivers +34.9% brief peak torque (242.8 N·m vs 180 N·m baseline) through phase-coordinated multi-rotor reaction torque summation. In the integrated Phoenix powertrain it provides a 140 kW brief burst and 118 kJ sprint reserve on a shared DC bus - bridging the gap between slow engine response and fast wheel demand. Hardware must validate efficiency, containment, and real boost duration before commercial claims.

Companion documents: ATPE Whitepaper v1.0; Project Phoenix Integrated System Whitepaper v1.0.